

Lab 8: Evaluate a Fraud Model and Alert Narrative

TGS-2026065050 · Tertiary Infotech Academy Pte Ltd

Lab 8: Evaluate a Fraud Model and Alert Narrative

Course: Generative AI for Finance and Fintech (TGS-2026065050)

Mapping: K3 · A4

Suggested time: 150 minutes

Objective: Choose a fraud threshold using precision, recall, business cost and explainability.

Scenario

A card issuer must reduce fraud loss without overwhelming investigators or blocking legitimate customers.

Files

- `northstar-fraud-risk-starter.xlsx` - learner working file
- `northstar-fraud-risk-solution.xlsx` - completed formulas, controls and charts for review
- `evidence-checklist.md` - submission and verification checklist

Before you begin

- Use only the supplied synthetic data.
- Do not enter live customer, payroll, account, credential or confidential data.
- If the required Microsoft feature is not enabled in your tenant, complete the workbook-based simulation and record the limitation.
- Keep facts, assumptions, AI suggestions and reviewer decisions visibly separate.

Detailed procedure

Step 1: Review the synthetic scored transactions and feature definitions.

Record the evidence for Step 1 in the checklist before continuing.

Step 2: Calculate confusion-matrix metrics at the supplied thresholds.

Record the evidence for Step 2 in the checklist before continuing.

Step 3: Estimate false-positive review cost and false-negative fraud loss.

Record the evidence for Step 3 in the checklist before continuing.

Step 4: Select a threshold that fits investigator capacity and risk appetite.

Record the evidence for Step 4 in the checklist before continuing.

Step 5: Explain three alerts using evidence and limitations.

Record the evidence for Step 5 in the checklist before continuing.

Step 6: Use GenAI to draft one case narrative, then verify and approve it as investigator work product.

Record the evidence for Step 6 in the checklist before continuing.

Prompt quality pattern

> Role: You are assisting a finance professional. Task: analyse only the named workbook table. Data boundary: do not use external or unstated facts. Controls: reconcile totals, identify missing evidence, and label assumptions. Output: return a concise table of finding, evidence, confidence, reviewer action and source cell or table.

Acceptance criteria

- The threshold decision cites precision, recall, cost and capacity; alert explanations use real fields; the generated narrative is labelled draft and evidence-linked.
- All generated claims have been checked against the workbook or approved knowledge.
- No live credentials, personal data or confidential business data appear in the evidence.
- A human reviewer and approval status are recorded.

Deliverable

A threshold decision, confusion matrix, alert queue and controlled case narrative.

Reflection

1. What did the AI accelerate?
2. What still required finance judgement?
3. Which control prevented the most serious failure?
4. What evidence is needed before this design can scale?

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